RESTRICTED DEFUSAL PROTOCOL

Free Fall: Defusal & Operations Manual

DOCUMENT REF: FF-MAN-REV4 TARGET UNIT: ELEVATOR EMERGENCY ESCAPE CONSOLE CLEARANCE: FIELD OPERATOR

MODULE 01: 3×3 ALPHANUMERIC MATRIX

KEYPAD DEFUSAL

Read instructions strictly from top to bottom. Match the FIRST applicable condition against the 6-character serial number, execute the designated action, and disregard all subsequent steps.

STEP 1: Numeric Suffix (Trailing Digits)

> **Condition: The last 3 characters of the serial number are all digits, AND there is at least one letter among the first 3 characters.**

- **A) Sum of the last 3 digits is ODD:**
  - If the 1st character of the serial number is a **Letter** → Press: `A1, B2, C3`
  - If the 1st character of the serial number is a **Digit** → Press: `A1, B1, B2, B3`
- **B) Sum of the last 3 digits is EVEN:**
  - If the 1st character of the serial number is a **Letter** → Press: `A1, A2, A3, B3`
  - If the 1st character of the serial number is a **Digit** → Press: `A1, B1, C1, C2, C3`

STEP 2: Numeric Prefix (Leading Digits)

> **Condition: The first 3 characters of the serial number are all digits, AND there is at least one letter among the last 3 characters.**

- **A) Sum of the first 3 digits is ODD:**
  - If the serial number contains **any Consonant** → Press: `A1, A2, B2, C2`
  - If the serial number contains **no Consonants** → Press: `A3, B2, C1`
- **B) Sum of the first 3 digits is EVEN:**
  - If the serial number contains **any Consonant** → Press: `A1, A3, B2, C1, C3`
  - If the serial number contains **no Consonants** → Press: `A1, A2, A3, B2, C3`

STEP 3: Emergency Fallback

> **Condition: The serial number satisfies neither Step 1 nor Step 2.**

Press every button on the matrix keypad → `A1, A2, A3, B1, B2, B3, C1, C2, C3`

MODULE 02: FREQUENCY REGULATOR (GHz)

DUAL-DIAL CALIBRATION

Read instructions strictly from top to bottom. Locate the first rule matching the displayed frequency number, tune to the indicated GHz frequency, and set Knob 1 (Left) and Knob 2 (Right).

STEP 1: Single-Digit Numbers (< 10)

> **Condition: The displayed value consists of only one single digit.**

Action: Set the exact displayed digit on **both knobs** (e.g. Value 1 → Left: `1` | Right: `1`).

STEP 2: Two-Digit Numbers (10 – 99)

> **Condition: The displayed number falls strictly within 10 to 99.**

- **A) Number is less than or equal to 50 (≤ 50):**
  • Divisible by 5 without remainder → Set dials to: `7` and `6`
  • Otherwise → Set dials to: `9` and `1`
- **B) Number is strictly greater than 50 (> 50):**
  • If there is AT LEAST 1 fuse present in the elevator → Set dials to: `8` and `9`
  • Otherwise (No fuses present) → Set dials to: `4` and `9`

STEP 3: Three-Digit Numbers (≥ 100)

> **Condition: The displayed frequency is greater than or equal to 100.**

- **A) The elevator contains MORE than 2 Puzzle Modules (> 2):**
  • If the number ends in 0 → Set **Hundreds digit** on Left Knob and **Tens digit** on Right Knob.
  • Otherwise → Set **Hundreds digit** on Left Knob and **Units (ones) digit** on Right Knob.
- **B) The elevator contains AT MOST 2 Puzzle Modules (≤ 2):**
  • If the number ends in 0 → Set Left Knob to `0` and Right Knob to `2`.
  • Otherwise → Set **Units (ones) digit** on Left Knob and **Tens digit** on Right Knob.

MODULE 03: COLOR & SHAPE SEQUENCE

SEQUENCE SELECTION (3 TILES)

Display shows 4 to 7 tiles. The objective is to assemble a sequence of exactly 3 tiles. Read sequentially from Step 1 to Step 3.

Cool Colors (C): Blue, Purple	**Warm Colors (W):** Yellow, Red	**Sharp Shapes:** Diamond (Square), Triangle	**Smooth Shapes:** Circle, Heart

STEP 1: No Dominant Color (Even Split / Exceptional State)

> **Condition: Neither Cool nor Warm colors hold a strict majority on the display.**

Arrange sequence in color pattern: `Cool - Warm - Cool (C W C)`

STEP 2: COOL Colors Dominate (Cool > Warm)

- **A) Total screen tiles: 5, 6, or 7:**
  • *Smooth shapes predominate:* If Yellow is present → 1 Yellow and 2 Cool tiles in any order. Otherwise → 3 Cool tiles.
  • *Sharp shapes predominate:* If exactly 2 Triangles exist → Place 2 Triangles anywhere in the sequence, with at least 1 warm color among the three. Otherwise → Center tile must be Cool; outer two tiles must be Sharp shapes.
- **B) Total screen tiles: exactly 4:**
  • *Smooth shapes predominate:* If exactly 2 Circles exist → First two slots must be Circles, containing at least 1 Cool color overall. Otherwise → Ensure at least two Smooth shapes in the sequence.
  • *Sharp shapes predominate:* If Red is present → Last position must be Red (others arbitrary). Otherwise → Pattern `Cool - Warm - Cool (C W C)`.

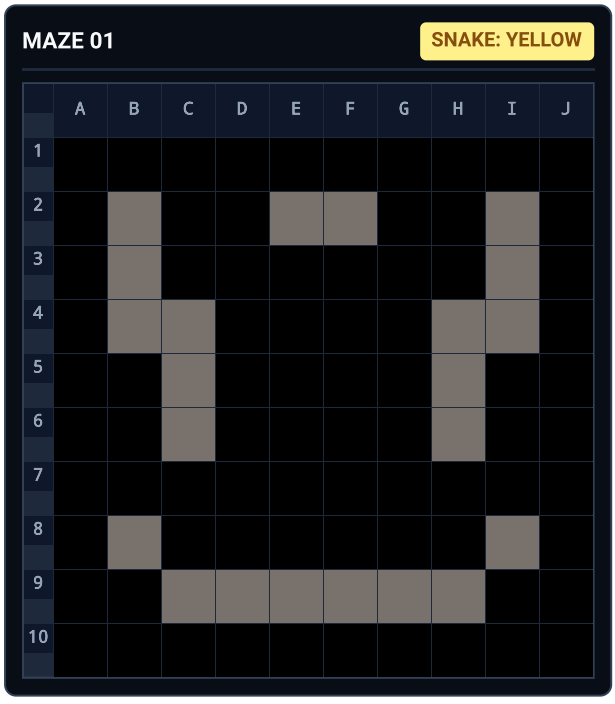
STEP 3: WARM Colors Dominate (Warm > Cool)

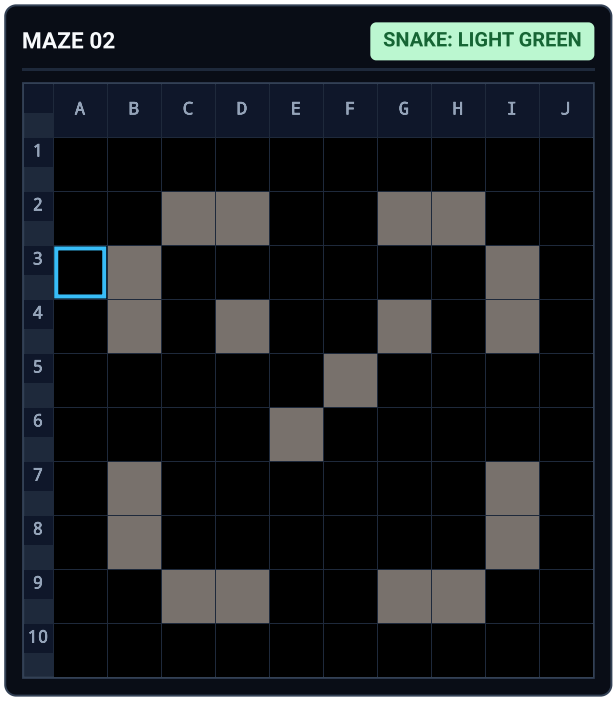
- **A) Total screen tiles: 5, 6, or 7:**
  • *Smooth shapes predominate:* If Blue is present → Place Blue in the last slot (others arbitrary). Otherwise → Pattern `Warm - Cool - Warm (W C W)`.
  • *Sharp shapes predominate:* If exactly 2 Diamonds exist → Place 2 Diamonds in slots 1 and 3 (middle tile arbitrary). Otherwise → Pattern `Warm - Warm - Cool (W W C)`.
- **B) Total screen tiles: exactly 4:**
  • *Smooth shapes predominate:* If exactly 2 Hearts exist → Place 2 Hearts in slots 2 and 3, with at least 1 Cool color overall. Otherwise → Pattern `Cool - Warm - Warm (C W W)`.
  • *Sharp shapes predominate:* If Purple is present → Pattern `Warm - Purple - Warm`. Otherwise → Select and place only Warm colors.

MODULE 04: SNAKE NAVIGATION (10×10 GRID)

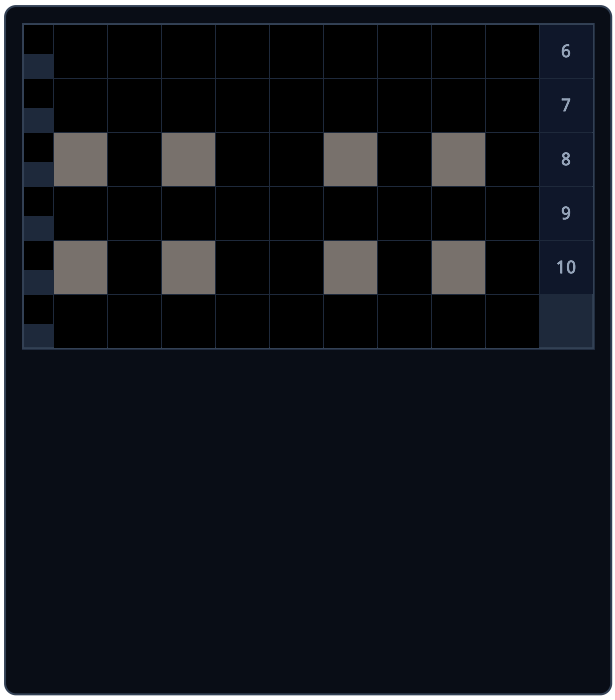
MAZE IDENTIFICATION BY COLOR

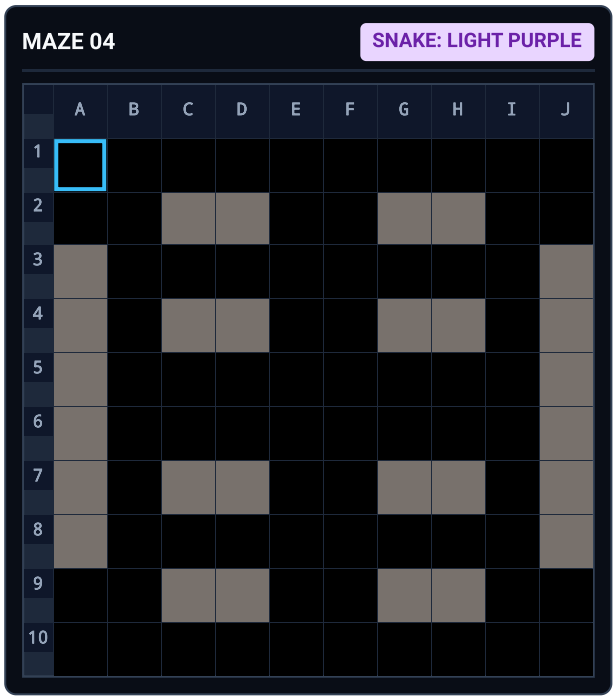
The display consists of a 10×10 grid (Columns A–J, Rows 1–10). Identify the maze by the player-controlled snake's color. Steer the snake to collect the target apple without colliding with gray barrier walls.

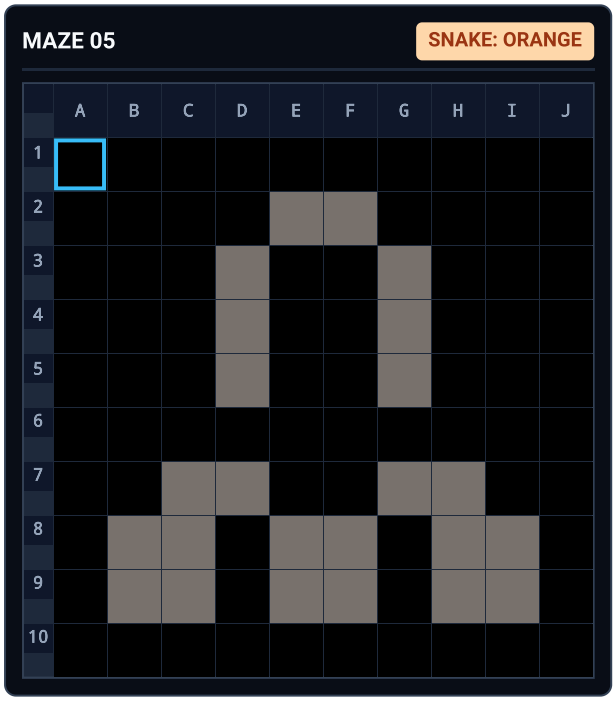


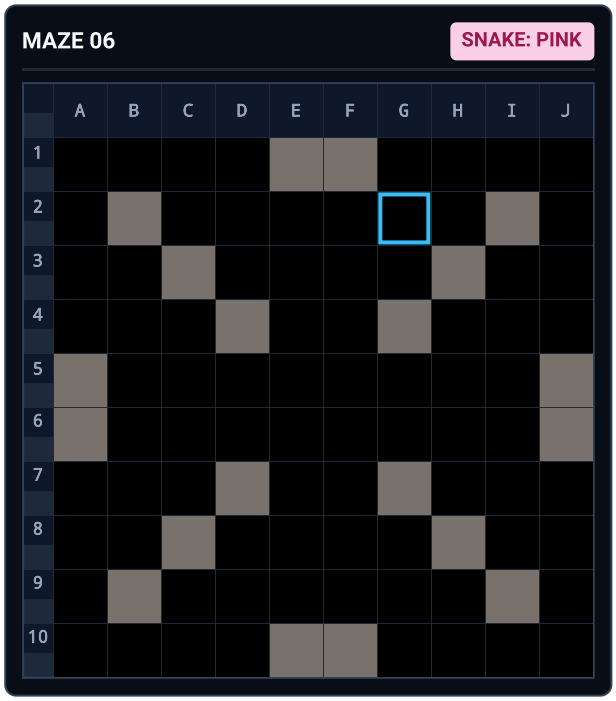


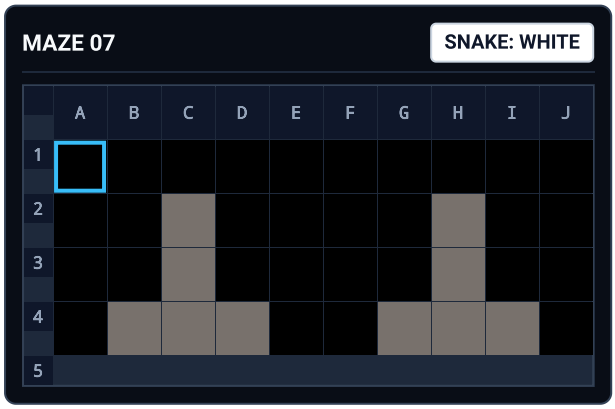


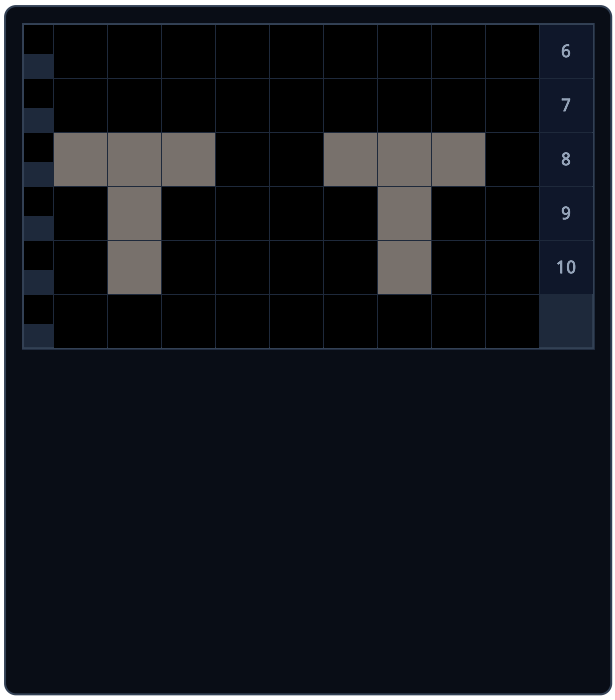

6
7
8
9
10

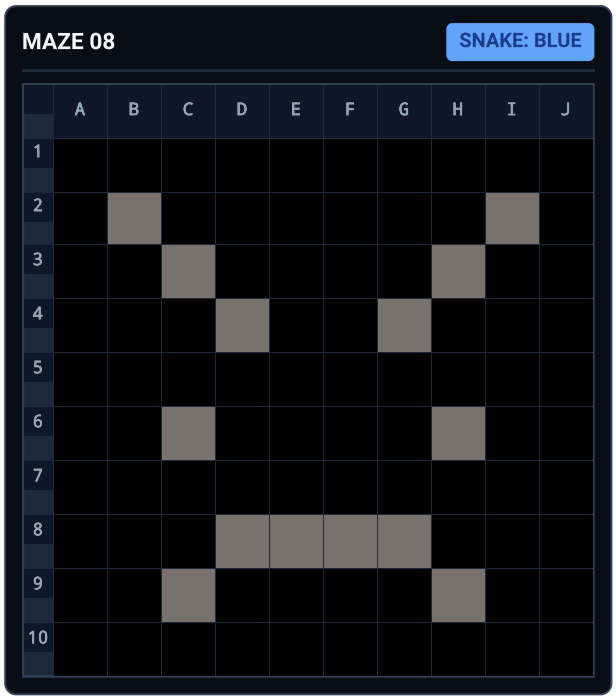

MAZE 04
SNAKE: LIGHT PURPLE
A B C D E F G H I J
1 2 3 4 5 6 7 8 9 10

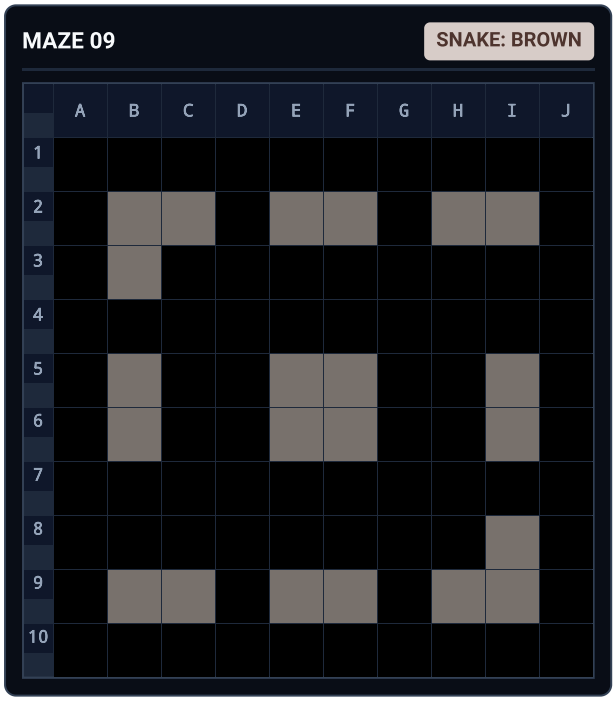

MAZE 05
SNAKE: ORANGE
A B C D E F G H I J
1 2 3 4 5 6 7 8 9 10

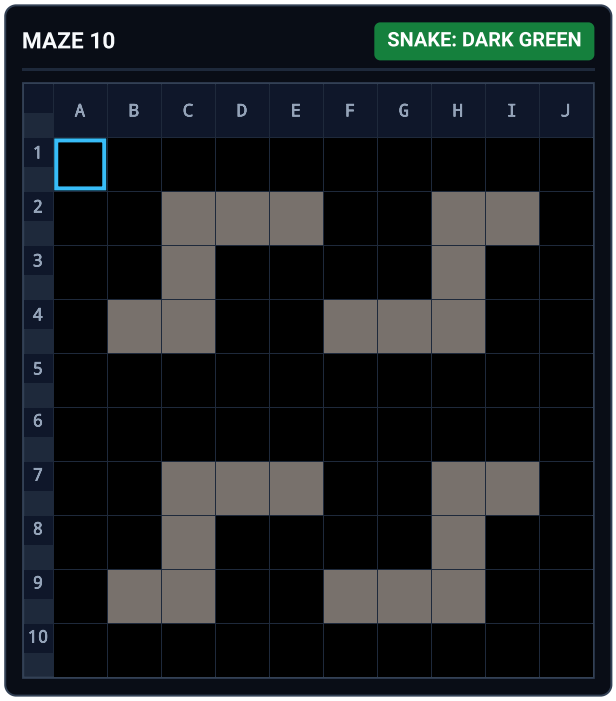

MAZE 06
SNAKE: PINK
A B C D E F G H I J
1 2 3 4 5 6 7 8 9 10


MAZE 07
SNAKE: WHITE
A B C D E F G H I J
1 2 3 4 5









MODULE 05: 7×7 GRID (PLAYER VS. AI)

TACTICAL LINE COMBAT

Consists of a 7×7 board and an active progress counter. The player competes against an AI algorithm to align their tokens into horizontal, vertical, or diagonal lines while navigating dynamic modifier tiles.

WIN AND DEFEAT CONDITIONS

The module is solved when the **Player is the first** to achieve any one of the following criteria:

- Form **3 lines of 3 tokens** in length.
- Form **2 lines of 4 tokens** in length.
- Form **1 line of 5 tokens** in length.

WARNING: If the AI algorithm fulfills any of the winning criteria before the player, the module records an immediate STRIKE / DEFEAT.

PROGRESS COUNTER & LINE BORDER PRIORITY

The digital display continuously tracks verified lines: `PLAYER - 3: x/3, 4: x/2, 5: x/1|AI - 3: x/3, 4: x/2, 5: x/1`. Validated lines are bordered on the physical board. When a token overlaps multiple lines, the highest priority border is displayed (`5 > 4 > 3`):

- **No Border:** No scored line formed (< 3 consecutive tokens).
- **Thin Border:** Validated line of 3 tokens.
- **Thick Border:** Validated line of 4 tokens.
- **Flashing / Glowing Border:** Validated line of 5 tokens.

CORE RULES & ROUND LIFECYCLE

- **Turn Lifecycle:** A single turn comprises one action by either party. A full round consists of 2 turns (Player Action + AI Response).
- **Adjacency:** Up to 8 surrounding cells (cardinals + diagonals).
- **Active Tile Pool:** Exactly 3 out of 6 special tile types are active each round, randomized on module reset.

SPECIAL MODIFIER TILES DIRECTORY

TILE COLOR	MODIFIER NAME	MECHANICAL RULE & BOARD BEHAVIOR
Green	**Required Target**	If active, at least 3 green tiles spawn. The player **must occupy at least one** green tile. Neglecting this requirement permanently blocks module defusal.
Red	**Access Denied**	Completely impassable for the player. Any attempt to claim this space is rejected with an invalid-input buzz.
Orange	**Cloning**	Claiming this cell instantly duplicates an identical mark onto an empty adjacent tile (if available).
Purple	**Fading**	Countdown initiates upon occupation. After 3 turns elapse, the purple tile and its mark dissolve, leaving a neutral empty space.
Teal / Cyan	**Moving Blocker**	Blocked to both player and AI. Following each move, one random teal tile displaces into a vacant neighboring slot.
Yellow	**Bomb**	Placement ignites a 3-turn fuse. Upon detonating, a 3×3 explosion (center + 8 neighboring cells) permanently incinerates all marks and active tiles.